

$$1.601 - V_B = \frac{AB}{t_B} = \frac{\sqrt{15^2 + 30^2} \text{ m}}{5.00 \text{ min}} = \frac{33.5 \text{ m}}{\text{min}} = 40.2 \text{ m/min}$$

$$V_C = \frac{X_{AC}}{t_C} = \frac{5 \text{ m}}{10 \text{ min}} = 0.5 \text{ m/min}$$

$$V_D = \frac{X_{AD}}{t_D} = \frac{\sqrt{60^2 + 30^2} \text{ m}}{15 \text{ min}} = \frac{67.1 \text{ m}}{\text{min}} = 4.47 \text{ m/min}$$

so $V_C < V_D < V_B$

the angle of V_C is θ_C , the angle of V_B is θ_B

$$\theta_C = 0^\circ, \tan \theta_B = \frac{30 \text{ m}}{15 \text{ m}} = 2, \text{ so } \theta_B = 63.44^\circ$$



$$67. h = \frac{1}{2}gt^2 \Rightarrow t = 0.639 \text{ s}$$

$$V_0 t = d \Rightarrow V_0 = \frac{d}{t} = \frac{15.7 \text{ m}}{0.639 \text{ s}} = 24.6 \text{ m/s}$$

$$a = \frac{V_0^2}{r} = \frac{(24.6 \text{ m/s})^2}{1.5 \text{ m}} = 399 \text{ m/s}^2$$

$$74. t = 1.8 \text{ min} = 0.3 \text{ h}, V_0 t = d \Rightarrow V_0 = \frac{d}{t} = \frac{18.3 \text{ km}}{0.3 \text{ h}} = 61 \text{ km/h}$$

$$N \theta = 20^\circ, V_{wx} = V_w \cos 20^\circ, V_{wy} = V_w \sin 20^\circ$$

$$V_{a-w} = \sqrt{(V_0 + V_{wy})^2 + V_{wx}^2} = 20.2 \text{ km/h}$$

$$V_x = V \cos \theta_0 = 19.15 \text{ m/s}, V_y = V \sin \theta_0 = 16.07 \text{ m/s}$$

$$V_x t = d \Rightarrow t = 1.15 \text{ s}$$

$$(a) h = V_y t - \frac{1}{2}gt^2 = 24.96 \text{ m}$$

(b) V_x doesn't change

$$(c) V_y = V_y - gt = 4.8 \text{ m/s}$$

(d) If it goes to the highest point:

$$V_y = gt_h \Rightarrow t_h = 1.64 \text{ s} > t = 1.15 \text{ s},$$

therefore, it has not passed the highest point on its trajectory.

$$74. t = 1.8 \text{ min} = 0.03 \text{ h}, V = 11.67 \text{ m/s}, d = 5.5 \text{ km} = 5500 \text{ m}$$



$$V_N = V \sin \theta = 3.99 \text{ m/s}$$

$$V_0 t = d \Rightarrow V_0 = \frac{d}{t} = \frac{18.3 \text{ km}}{0.3 \text{ h}} = 61 \text{ km/h}$$

$$V_a = V - V_N (V_0 + V) \sin \theta - V_N \Rightarrow V_a = 49.0 \text{ km/h}$$